



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	-3.4653	Longitude	-62.2159
Zone size (km)	10	Analysis date	2026-10-01

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FOREST INTEGRITY SCORE / 100 Good Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.842	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.025	Stable
10-year trend (Landsat 8/9)	+0.456	Improving
40-year trend (Landsat 1984)	+0.493	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+1.23% / year	Amelioration
NDVI change 5 years	+0.007	Stable
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Old-growth forest (>120 years)	Ancient forest, high biodiversity
Above-ground biomass	~118.5 t/ha	Medium biomass (50-150 t/ha)
Canopy Height (GEDI+S2 10m)	13.2m max / 7.3m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	2,023,190 t CO ₂	CO ₂ sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO ₂ per hectare	257.6 t CO ₂ /ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 30,347,850	Estimated value at EUR 15/tonne CO ₂ (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO ₂ sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 2,528,987 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDi Tree Cover	100%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Dense Tropical Forest	High biodiversity, dense canopy
Deforestation Risk (AI)	Moderate	Moderate risk
Main cause (AI)	Anthropogenic deforestation, likely from land conversion for agriculture, cattle ranching, or settlement expansion typical of the Solimões River region in Amazonas state.	
Degradation signs (AI)	Recent deforestation detected in 4.8% of the pixel classification (2021-2026 period), indicating active forest loss. Low NDWI (-0.745) suggests water stress or dry season conditions. The deforested patches represent recent clearing activity in this region.	
AI Recommendation	Implement immediate monitoring and enforcement in the 4.8% deforested areas using high-resolution satellite imagery to prevent further expansion, while establishing forest protection corridors in the remaining 93.6% healthy forest to maintain ecosystem connectivity.	
Sentinel-1 Radar (VV)	-7.805 dB	Dense forest (radar)
Forest cover	99.9%	Excellent
Fire pixels detected (FIRMS/year)	146 pixels	~36 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	

Deforestation Alert	STABLE — No significant change	
Area change (3 years)	+3 ha	+0.03%
Annual rate (area/yr)	+1 ha/yr	Recovery
Area change (10 years)	+9060.5 ha	+988.7%
Annual rate 10yr (area)	+906 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	25	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	15	15
Fire score (0=many fires, 10=none)	0	10
Base score (satellite)	80	100
RF deforestation penalty (2021-2026)	-3	
FINAL SCORE (after RF correction)	77	100
AI Deforestation Risk	Moderate	

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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